



NORTHERN WOLVES
AIR CONDITIONING

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Customer
Callahan Construction Managers

Project: **Sunrise Rye Brook - Independent Living / Assisted Living**

Address: **900 King St, Rye Brook, NY 10573**

Mechanical drawings dated: 07.15.2026 (GMP set)

Mechanical Value Engineering Proposals

Northern Wolves reviewed the GMP mechanical set and proposes the value engineering items below. The design today relies on a 5,700 MBH boiler plant with hot water and glycol distribution, hydronic VAV reheat, fan coils and cabinet heaters, plus a full BMS. Most of this can be replaced by the gas rooftop units and heat pump DOAS already in the design and small electric or heat pump terminals, as done on other recent Sunrise communities. Pricing of each item will follow once the owner selects the items to pursue. Pool, spa and sauna systems are by others and not part of this review.

VE	Item	Area	Approvals
VE-01	Heating plant: delete boilers and hot water / glycol systems	AL basement boiler room, chimneys to roof, hot water and glycol piping throughout	Requires VE-02 to VE-06. IMEG redesign; Sunrise approval.
VE-02	VAV reheat: electric reheat only where needed	1st floor common areas, IL and AL (VAV-1 to VAV-47)	IMEG redesign and energy code check (VAV minimums); electrical service capacity by EE.
VE-03	Common areas: cancel VAV boxes in open areas and circulation	1st floor lounges, bistros, dining rooms, reception, corridors, kitchen, restrooms	IMEG zoning review (rooftop unit control zone per unit); Sunrise approval.
VE-04	Cabinet heaters: hot water to electric	Entries, vestibules and trash room (CH-1, CH-3 to CH-12)	IMEG; EE.
VE-05	Basement fan coil units: electric unit heaters	IL and AL basement back-of-house rooms (FCU-A)	IMEG; fire protection designer to confirm freeze-protection locations.
VE-06	Garage ramp snow melt	North and south garage ramps (HX-3, P-3 / P-4, 120 F glycol)	Owner decision; civil / landscape for drainage.
VE-07	Fire / smoke dampers at apartment duct branches	Floors 1 to 4, IL and AL corridors (about 360 FSD tags)	IMEG / HKS code analysis (AL Group I-1 smoke barriers), building department.
VE-08	Apartment bath exhaust to the DOAS energy recovery	All apartments, IL and AL (TX-A)	IMEG; DOAS exhaust airflow check.
VE-09	Controls: standalone equipment controls in lieu of BMS	Building-wide	Sunrise approval of the controls standard.

VE-01 requires VE-02 to VE-06. Electrical work for electric heat by EC; electrical service capacity to be confirmed by the EE and utility.

VE-01 Heating plant: delete boilers and hot water / glycol systems

Area: AL basement boiler room, chimneys to roof, hot water and glycol piping throughout

As designed

- Three 1,999 MBH gas boilers B-1, B-2, B-3 (5,700 MBH output), primary / secondary with HS-1
- Pumps BP-1/2, PP-1/2, P-1 to P-4, expansion tanks ET-1/2, air separator AS-1, heat exchanger HX-3 to 30% glycol
- Fire-rated chimney enclosures basement to roof, combustion air areaway, water treatment, hot water and glycol distribution to the 1st floor and basement
- Connected heating load about 2,430 MBH: the plant is about 2.3 times the load

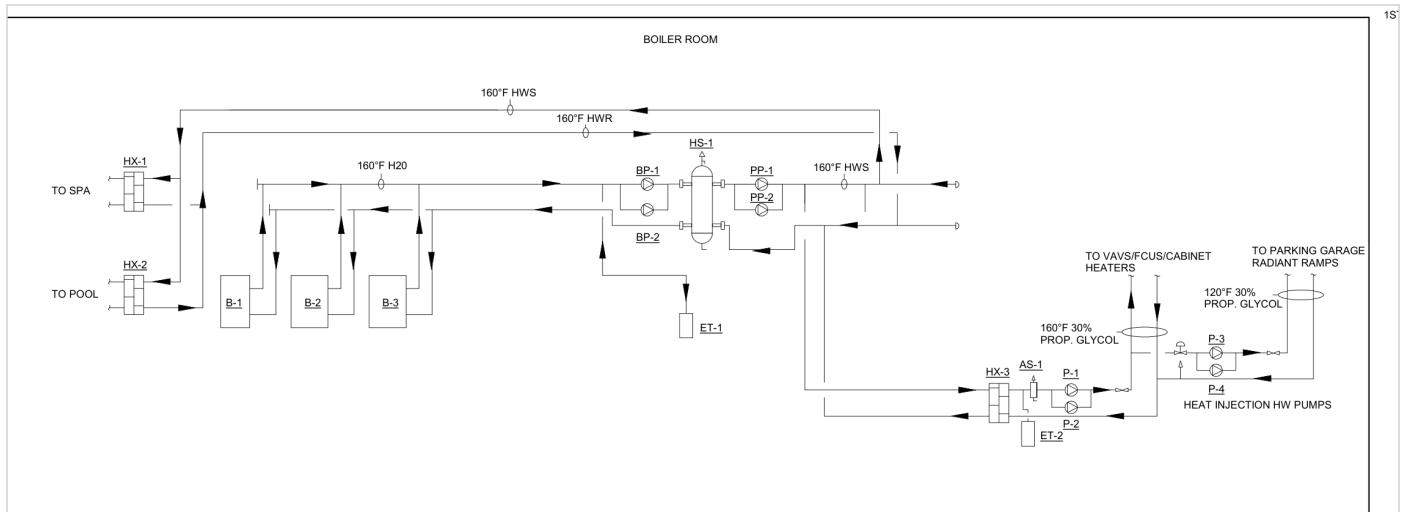
Proposed

- Delete the boiler plant and all hot water / glycol distribution; the building runs on gas rooftop units, heat pump DOAS and the electric terminal heat in VE-02 to VE-06
- No boiler room equipment, chimneys, combustion air, water treatment, flushing or hydronic balancing
- Pool and spa heating by the pool contractor (by others)

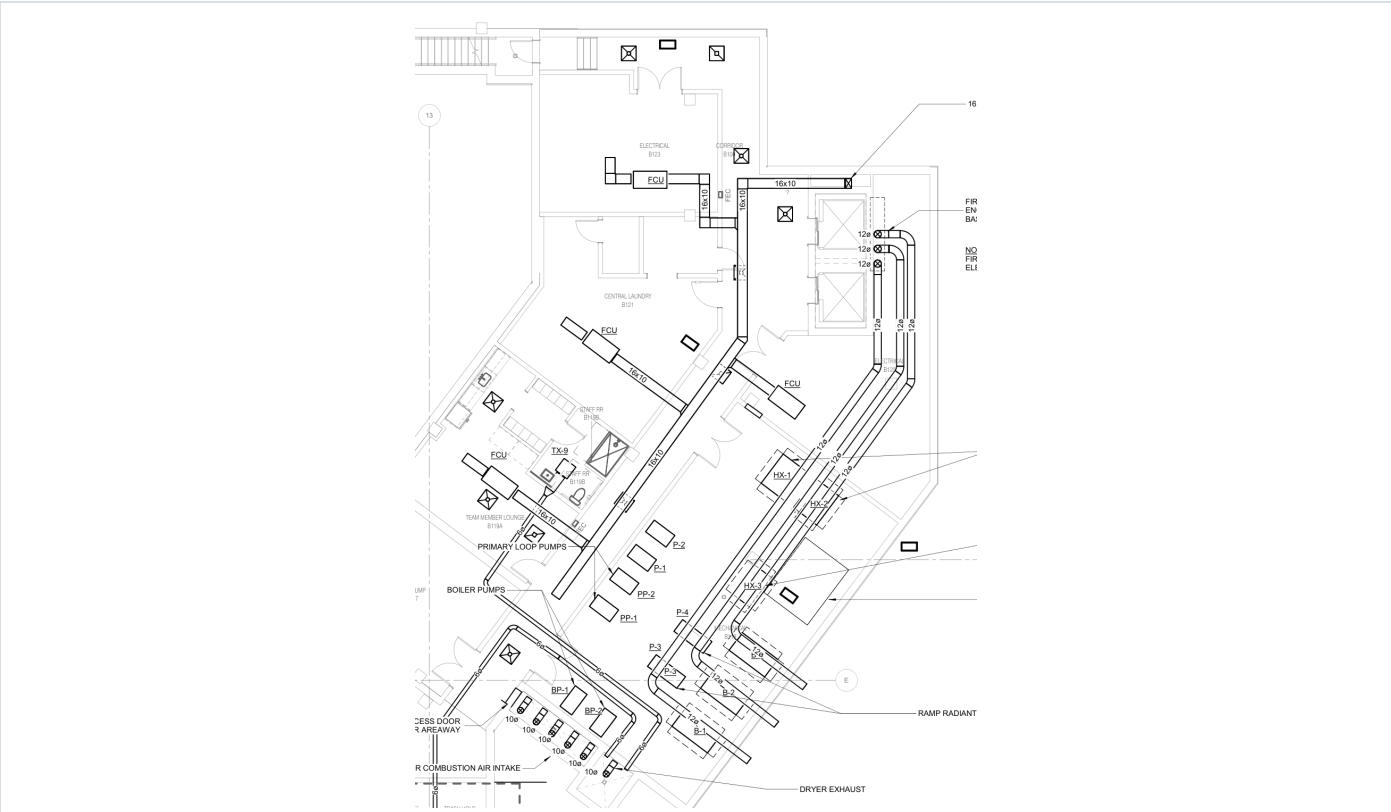
Benefits: Largest single reduction in mechanical, controls and GC scope (chimney shafts, areaway, boiler room fit-out). Removes the boiler plant maintenance and its BMS sequences.

Electrical impact: See VE-02 to VE-05

Approvals / coordination: Requires VE-02 to VE-06. IMEG redesign; Sunrise approval.



M5000 Hydronic flow diagram - boiler room



M2002 AL basement - boiler room, pumps, heat exchangers, chimney enclosures

BOILER SCHEDULE - HOT WATER (BLR)										
NOTES: 1.										
TAG NAME	SERVICE	FUEL		MAXIMUM DIMENSIONS			WEIGHT LBS	MANUFACTURER	MODEL	NOTES
		TYPE	MBH	WIDTH	HEIGHT	DEPTH				
B-1	BUILDING HEATING	FIRE TUBE	1999	1800	3'-0"	6'-8"	1,750	LAARS MAGNITHERM FT	2000	
B-2	BUILDING HEATING	FIRE TUBE	1999	1800	3'-0"	6'-8"	1,750	LAARS MAGNITHERM FT	2000	
B-3	BUILDING HEATING	FIRE TUBE	1999	1800	3'-0"	6'-8"	1,750	LAARS MAGNITHERM FT	2000	

PUMP SCHEDULE (P)							
NOTES: 1.							
TAG NAME	LOCATION	SERVICE	GPM DESIGN	FT. HEAD	WEIGHT LBS		MANUFACTURER AND MODEL
					OPERATING		
BP-1	AL BASEMENT	BOILER	381	30	141		ARMSTRONG 4300 0610-005
BP-2	AL BASEMENT	BOILER	381	30	141		ARMSTRONG 4300 0610-005
P-1	AL BASEMENT	HX-3	55	40	67		ARMSTRONG 4300 1205-001.5
P-2	AL BASEMENT	HX-3	55	40	67		ARMSTRONG 4300 1205-001.5
P-3	AL BASEMENT	HX-3	56	30	67		ARMSTRONG 4300 1205-000.7
P-4	AL BASEMENT	HX-3	56	30	67		ARMSTRONG 4300 1205-000.7
PP-1	AL BASEMENT	BOILER	0	0	0		

M6001 Boiler and pump schedules

VE-02 VAV reheat: electric reheat only where needed

Area: 1st floor common areas, IL and AL (VAV-1 to VAV-47)

As designed

- 47 VAV boxes with hot water reheat coils (629 MBH, 45 GPM) on 5 rooftop units
- Hot water piping, control valves, balancing and insulation to every box (M2014 to M2016)

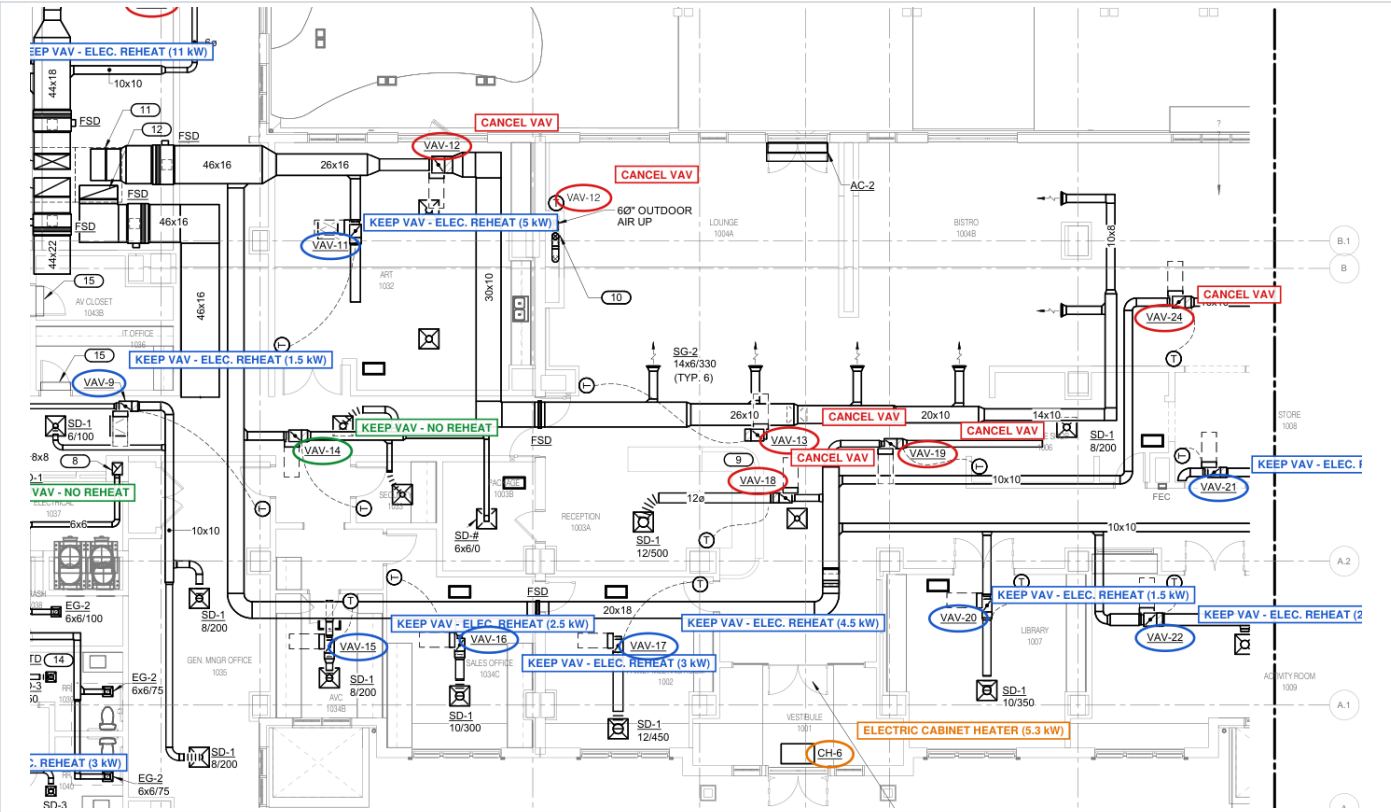
Proposed

- Same rooftop units as designed (gas heat). VAV boxes kept in enclosed rooms (VE-03) get factory electric reheat coils with SCR control and a simple standalone thermostat: 25 boxes
- Interior / back-of-house rooms: VAV box kept cooling-only, no reheat required since the rooftop units are gas heat (VAV-8, VAV-14, VAV-31, VAV-32)
- No hot water piping, coil packages or control valves at any box. See mark-ups M2010, M2011, M2013

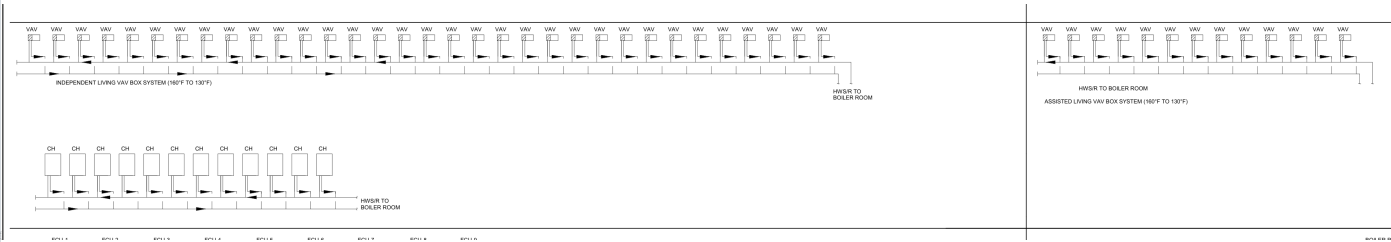
Benefits: Deletes 1st floor hot water piping and 47 valve / coil packages; standalone thermostats instead of BMS zone control.

Electrical impact: About +97 kW connected (25 boxes, per-box kW on the mark-ups: scheduled HW MBH / 3.412), versus about 185 kW if all 47 were electric

Approvals / coordination: IMEG redesign and energy code check (VAV minimums); electrical service capacity by EE.



Mark-up M2010 IL area A - reception / lounge / bistro and offices



M5000 Hydronic flow diagram - IL and AL VAV reheat and cabinet heaters

VAV SCHEDULE - HYDRONIC REHEAT																									
TAG	AHU	ROOM	MODEL	SIZE		CFM		STATIC PRESSURE				NE LEVELS		HOT WATER HEATING COIL										UNIT INFO	
				UNIT	OUTLET	MAX	MIN	INLET	DOWN	MIN	RAD	DIS	CFM	MIN	EAT	ENT	LAT	APV	GPM	LWT	WPU	ROW	PP		HAND
VAV-01		DESV	12	16x15	1200	625	1	0.25	0.24	20	24	825	20.9	60	160	90	0.23	1.5	122.5	0.24	2-RH	10	RH		
VAV-02		DESV	14	20x17.5	1400	1350	1	0.25	0.19	20	19	1350	44	60	160	90	0.17	2.4	122.6	0.3	2-RH	10	RH		
VAV-03		DESV	14	20x17.5	2250	1150	1	0.25	0.38	23	22	1150	37.5	60	160	90	0.26	1.9	120	0.22	2-RH	10	RH		
VAV-04		DESV	08	12x10	700	375	1	0.25	0.22	22	26	375	12.2	60	160	90	0.3	0.8	127.1	0.10	2-RH	10	RH		
VAV-05		DESV	08	12x10	400	400	1	0.25	0.13	17	25	400	13	60	160	90	0.12	0.8	127.6	0.17	2-RH	10	RH		
VAV-06		DESV	08	12x10	400	400	1	0.25	0.13	17	25	400	13	60	160	90	0.12	0.8	127.6	0.17	2-RH	10	RH		
VAV-07		DESV	10	14x12.5	700	250	1	0.25	0.09	23	27	250	8.1	60	160	90	0.08	1	143.9	0.15	1-RH	10	RH		
VAV-08		DESV	04	12x8	200	100	1	0.25	0.11	27	34	100	3.3	60	160	90	0.03	0.4	140.8	0.1	1-RH	10	RH		
VAV-09		DESV	08	12x10	350	150	1	0.25	0.06	17	25	150	4.9	60	160	90	0.05	0.5	137.6	0.19	1-RH	10	RH		
VAV-10		DESV	08	12x8	425	250	1	0.25	0.38	23	24	250	8.9	60	160	90	0.21	0.7	134.3	0.1	2-RH	10	RH		
VAV-11		DESV	10	14x12.5	300	500	1	0.25	0.12	26	24	500	16.6	60	160	90	0.11	1.1	127.6	0.14	2-RH	10	RH		
VAV-12		DESV	14	20x17.5	1300	1300	1	0.25	0.19	19	19	1300	43.3	60	160	90	0.14	2.3	123.3	0.28	2-RH	10	RH		
VAV-13		DESV	10	14x12.5	650	650	1	0.25	0.14	30	24	650	19.5	60	160	90	0.13	1.3	128.7	0.18	2-RH	10	RH		
VAV-14		DESV	10	14x17.5	400	400	1	0.25	0.14	30	34	400	19.4	60	160	90	0.13	1.3	128.7	0.18	2-RH	10	RH		

M6000 VAV schedule - hydronic reheat (partial)

VE-03 Common areas: cancel VAV boxes in open areas and circulation

Area: 1st floor lounges, bistros, dining rooms, reception, corridors, kitchen, restrooms

As designed

- 47 VAV zones on 5 rooftop units, including open areas that are one space with their neighbors

Proposed

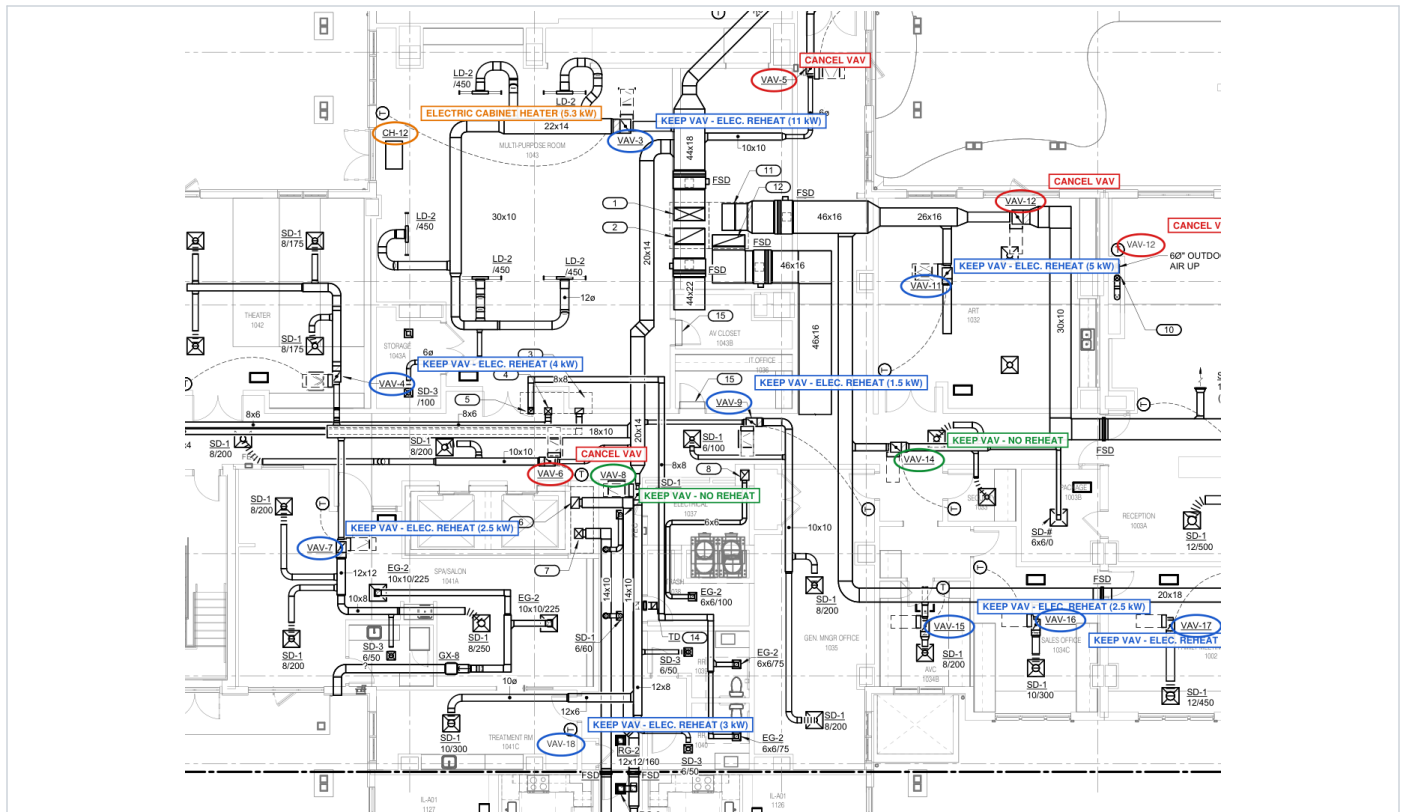
- Same rooftop units and main ductwork as designed. Cancel 18 VAV boxes in open areas and circulation (lounge, bistro, coffee shop, reception, formal / casual / AL dining, parlor, kitchen, corridors, restrooms)
- Branch ducts stay, with manual volume dampers balanced to the design maximum CFM; the area is served directly by its rooftop unit (rooftop unit space sensor in that area), gas heat from the rooftop unit
- VAV boxes stay only in enclosed rooms (offices, theaters, activity, fitness, salon, private dining, library, family rooms) - see VE-02. No ductless units added

- Each box is marked on the attached mark-ups: CANCEL VAV / KEEP VAV - ELEC. REHEAT / KEEP VAV - NO REHEAT

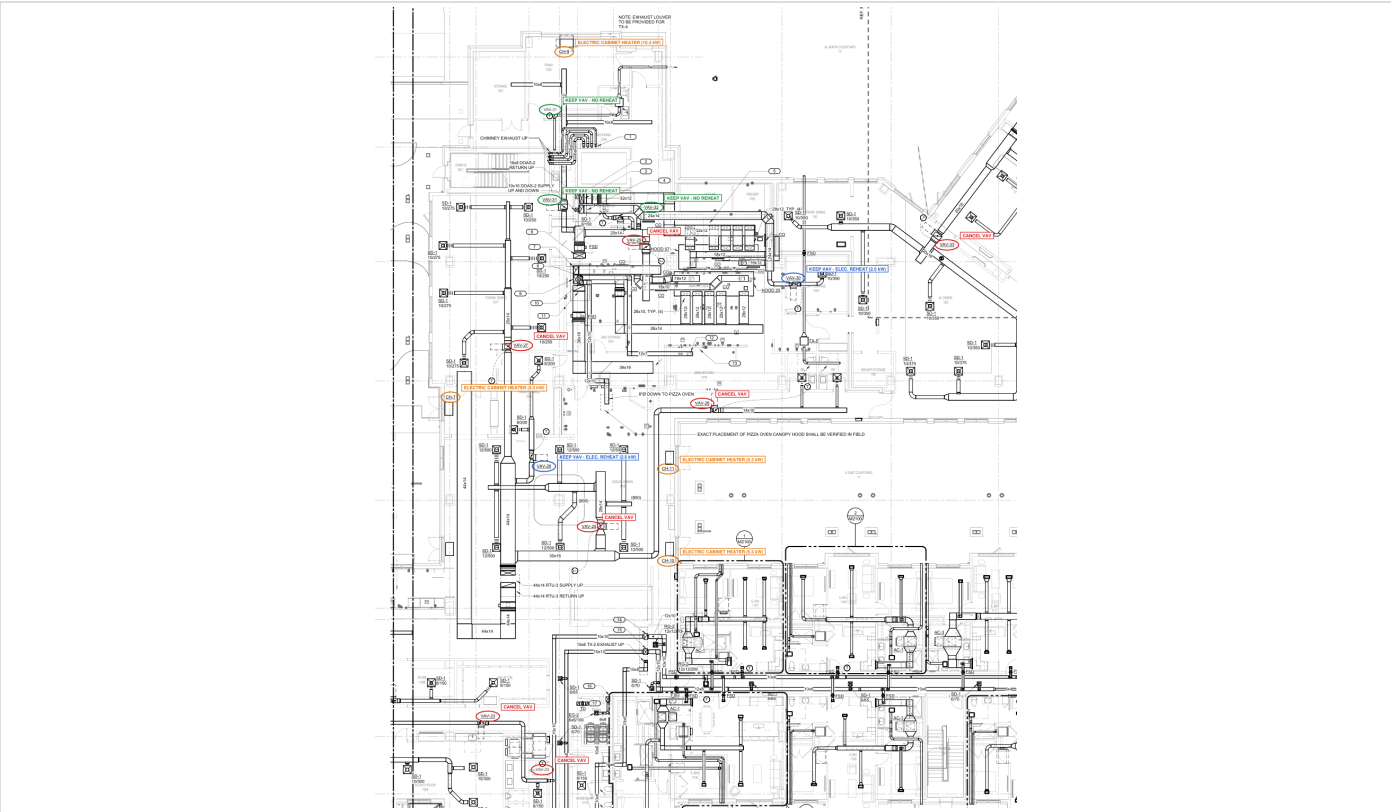
Benefits: 18 fewer VAV boxes, reheat coils, thermostats and controls; simpler operation.

Electrical impact: Reduces electric reheat (included in VE-02 figure)

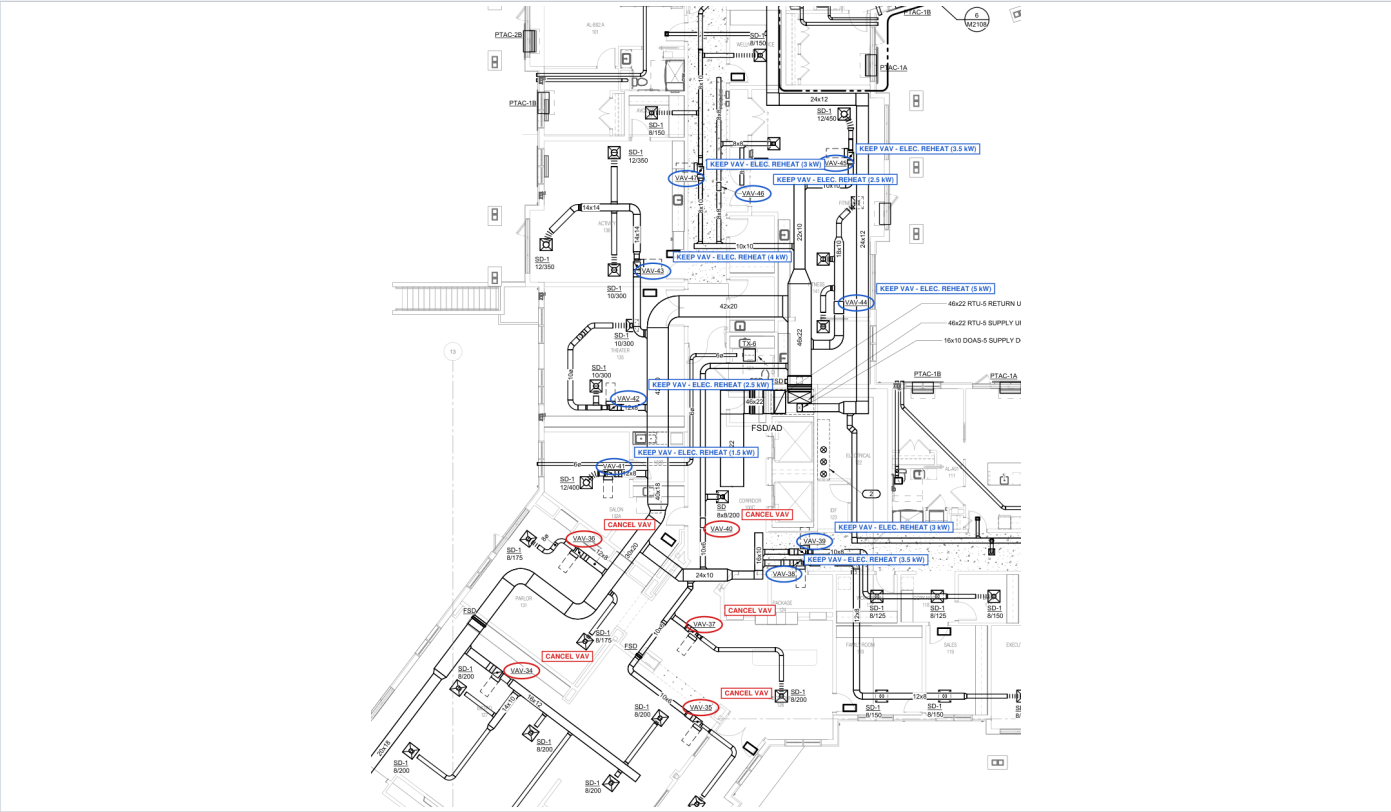
Approvals / coordination: IMEG zoning review (rooftop unit control zone per unit); Sunrise approval.



Mark-up M2010 IL area A - multi-purpose, theater, salon, offices, corridor



Mark-up M2011 IL area B - dining rooms, kitchen, restrooms



Mark-up M2013 AL area D - bistro, parlor, reception, corridor, activity rooms

VE-04 Cabinet heaters: hot water to electric

Area: Entries, vestibules and trash room (CH-1, CH-3 to CH-12)

As designed

- 11 hot water cabinet heaters in the schedule (CH-8 not shown on plans) with piping and BMS control

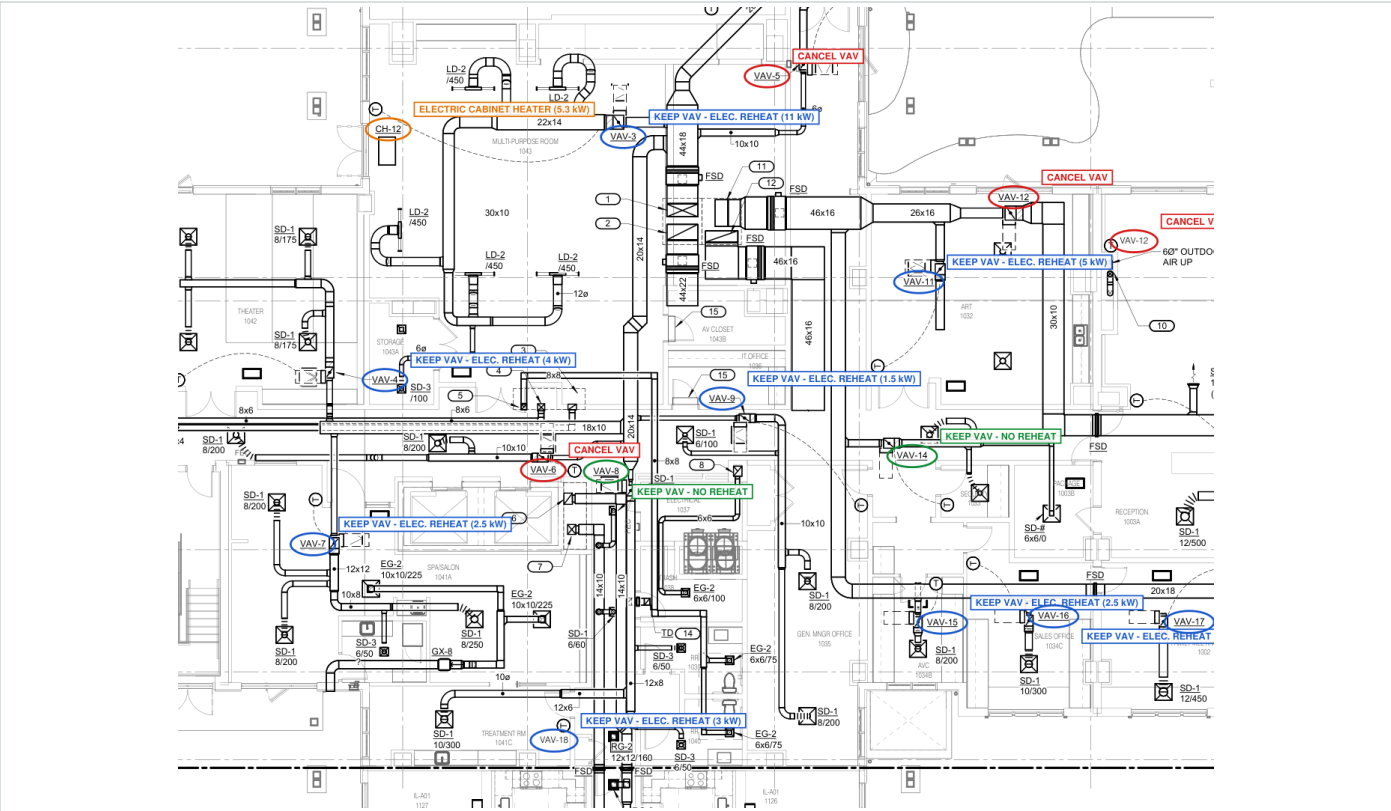
Proposed

- Electric cabinet heaters with integral thermostats at the same locations; no piping, no BMS points. Each heater is marked on mark-ups M2010 and M2011

Benefits: Deletes piping to 11 remote locations; standalone control.

Electrical impact: About +58 kW connected for the 10 heaters on the plans (5.3 kW typical, CH-9 10.4 kW)

Approvals / coordination: IMEG; EE.



Mark-up M2010 - CH-12 and other entries

CABINET HEATER SCHEDULE																			
NOTES: 1.COORDINATE COLOR SELECTION WITH ARCHITECT.																			
TAG NAME	LOCATION	SERVICE	DIMENSIONS			FAN		COIL	GPM	COIL		ELECTRIC AL		MAXIMUM DIMENSIONS			WEIGHT LBS	MANUFACTURER	MO
	NAME		WIDTH	HEIGHT	DEPTH	CFM DESIGN	DRIVE			MBH	VOLTS	PH	WIDTH	HEIGHT	DEPTH	DRY			
CH-1		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105	MODINE	CW0	
CH-3		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105	MODINE	CW0	
CH-4		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105	MODINE	CW0	
CH-5		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105	MODINE	CW0	
CH-6		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105	MODINE	CW0	
CH-7		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105	MODINE	CW0	
CH-8		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105	MODINE	CW0	
CH-9		BOILER	40"	12"	24"	1,050	CENTRIFUGAL	HOT WATER	5.8	35552	120	1	5'-11 3/4"	2'-4"	1'-0"	185	MODINE	CW0	
CH-10		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105	MODINE	CW0	
CH-11		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105	MODINE	CW0	

M6001 Cabinet heater schedule

VE-05 Basement fan coil units: electric unit heaters

Area: IL and AL basement back-of-house rooms (FCU-A)

As designed

- FCU-A, quantity 20, "basement heating", hot water / glycol heating coil only (schedule data blank); M5000 shows FCU-1 to FCU-17 on the glycol loop with RAD-1/2
- Each FCU takes a small DOAS branch at its inlet and recirculates the room
- No glycol or hot water piping to these units is drawn on M2001 / M2003

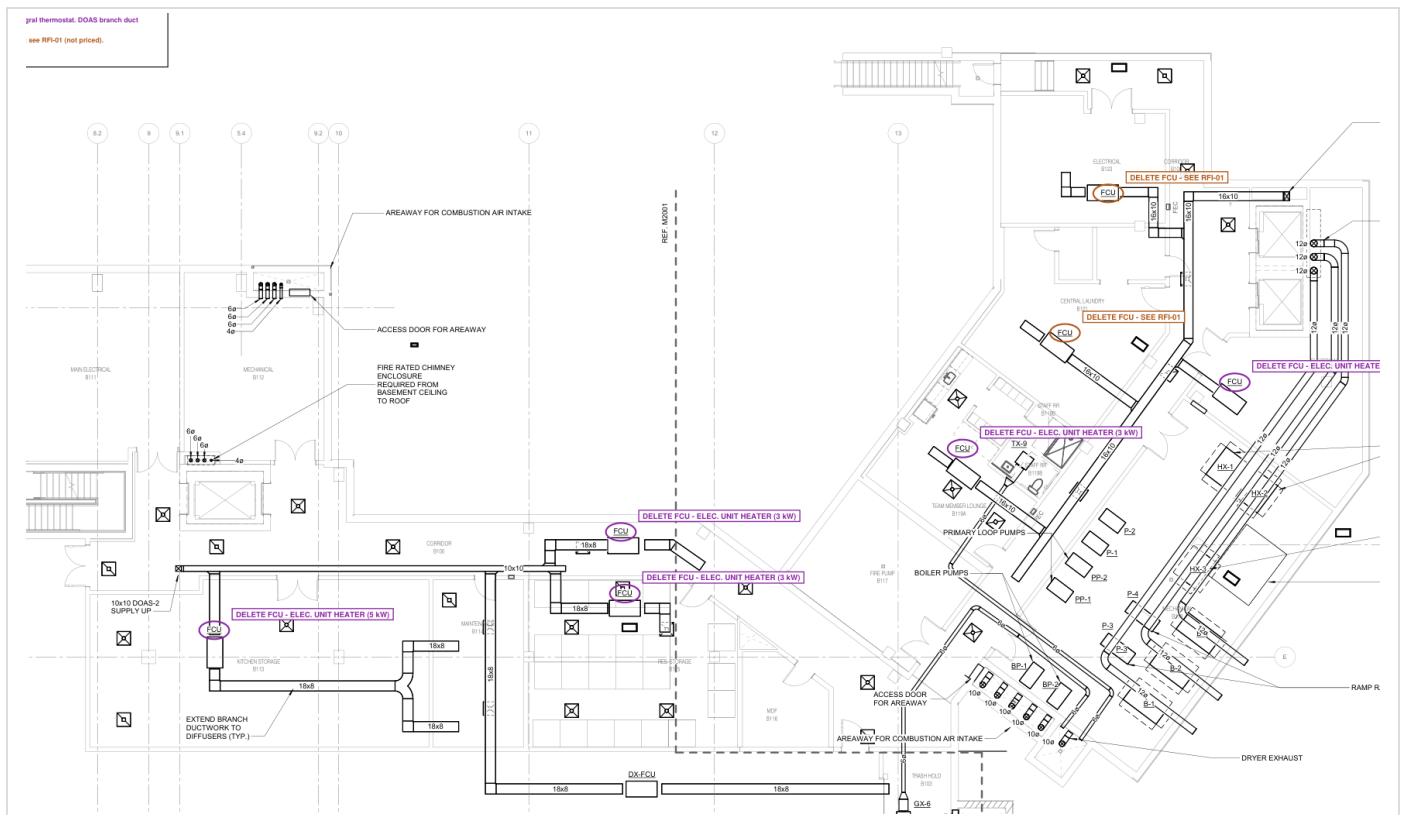
Proposed

- Delete the FCUs and RAD-1/2. Electric unit heater with integral thermostat in each room that had an FCU (storage, trash, corridor, staff lounge, mechanical); DOAS branch duct extended directly to the room diffuser, return grille deleted. Each unit is marked on mark-ups M2000 and M2002
- Electrical room B123 and central laundry B121: FCU deleted; these rooms need cooling / ventilation, not heat - see RFI-01 (separate page, not priced)

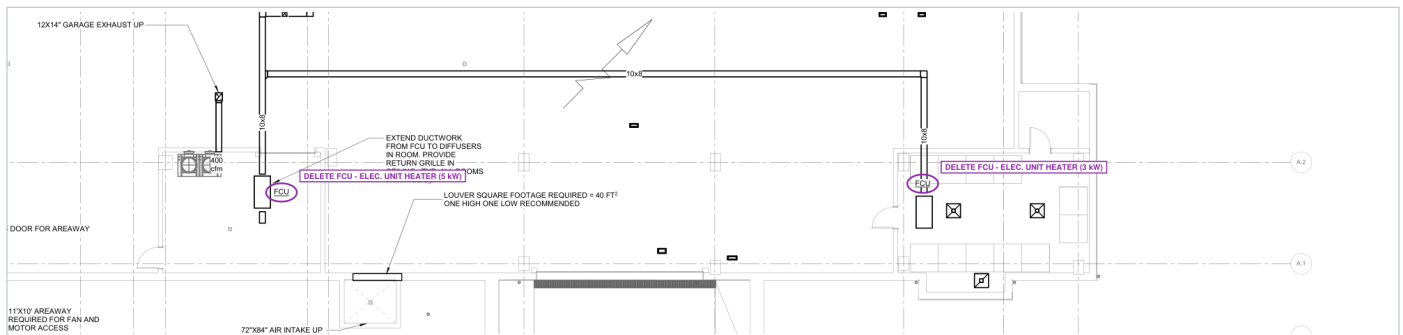
Benefits: Deletes the fan coils, glycol piping and their controls.

Electrical impact: About +35 kW connected (3 kW storage / corridor / lounge, 5 kW larger rooms and trash rooms; per-room kW on the mark-ups)

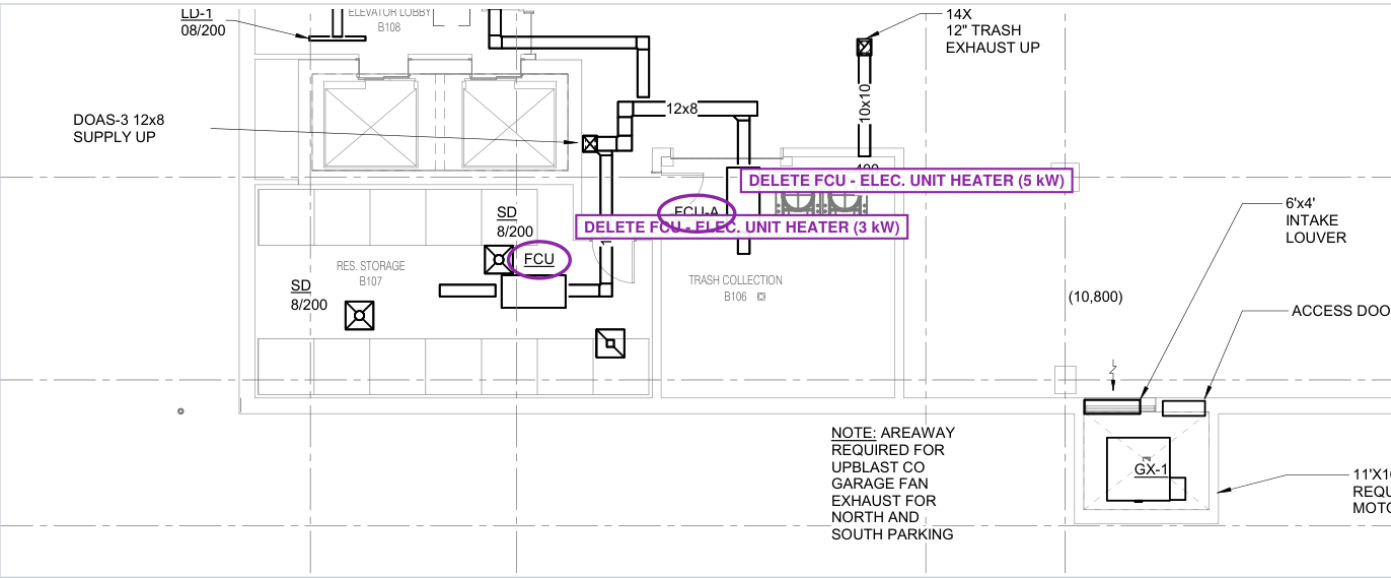
Approvals / coordination: IMEG; fire protection designer to confirm freeze-protection locations.



Mark-up M2002 AL basement - storage, corridor, staff lounge, mechanical, electrical, laundry



Mark-up M2002 AL basement south - trash and storage rooms



Mark-up M2000 IL basement - residents' storage, trash collection

FAN COIL UNIT SCHEDULE - HYDRONIC (FCU)																						
NOTES: 1.																						
TAG NAME	QUANTITY	SERVICE	TYPE	FAN			HEATING COIL							ELECTRICAL		CONTROL		MAXIMUM DIMENSIONS			WEIGHT	
				CFM DESIGN	EXT S.P. IN. WC	MOTOR MHP [E]	MBH	EAT DB °F	LAT DB °F	GPM DESIGN	W.P.D. FT. HD.	EWL °F	LWT °F	VOLTS	PH	TYPE [H]	DIAGRAM	WIDTH	HEIGHT	DEPTH	DRY	MANUF
FCU-A	20	BASEMENT HEATING	HORIZONTAL	0	0	0	0	0	0	0	0	0	0	0	1			0"	0"	0"	0	

M6000 Fan coil unit schedule (FCU-A, quantity 20, no data)

VE-06 Garage ramp snow melt

Area: North and south garage ramps (HX-3, P-3 / P-4, 120 F glycol)

As designed

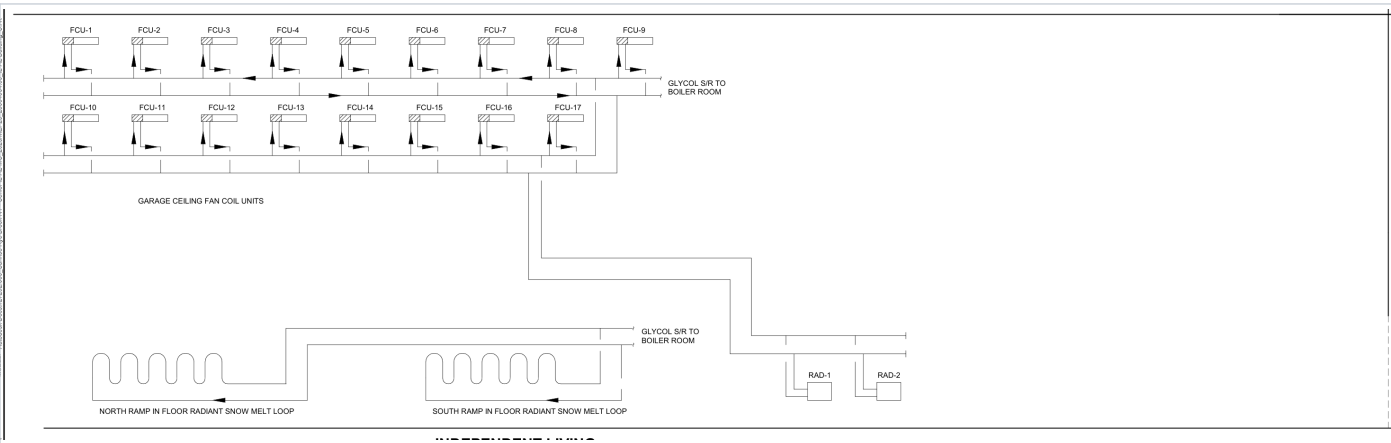
- In-floor radiant snow melt loops on both ramps from the boiler plant through HX-3

Proposed

- Option A: no snow melt; trench drains and salting (owner decision)
 - Option B: electric snow melt cable by EC, storm-only operation with load shedding
- Benefits: Removes the long glycol mains from the boiler room to the ramps and the snow melt pumps.

Electrical impact: Option A: 0; option B: about +240 kW (storm only)

Approvals / coordination: Owner decision; civil / landscape for drainage.



M5000 Hydronic flow diagram - basement FCUs, RAD-1/2 and ramp snow melt loops

VE-07 Fire / smoke dampers at apartment duct branches

Area: Floors 1 to 4, IL and AL corridors (about 360 FSD tags)

As designed

- Combination fire / smoke damper at every DOAS supply / return branch into each apartment (M2023 note: "provide fire/smoke damper at all DOAS supply/return penetrations through demising walls")
- Each damper needs power, fire alarm interface, access door in a finished ceiling and periodic testing

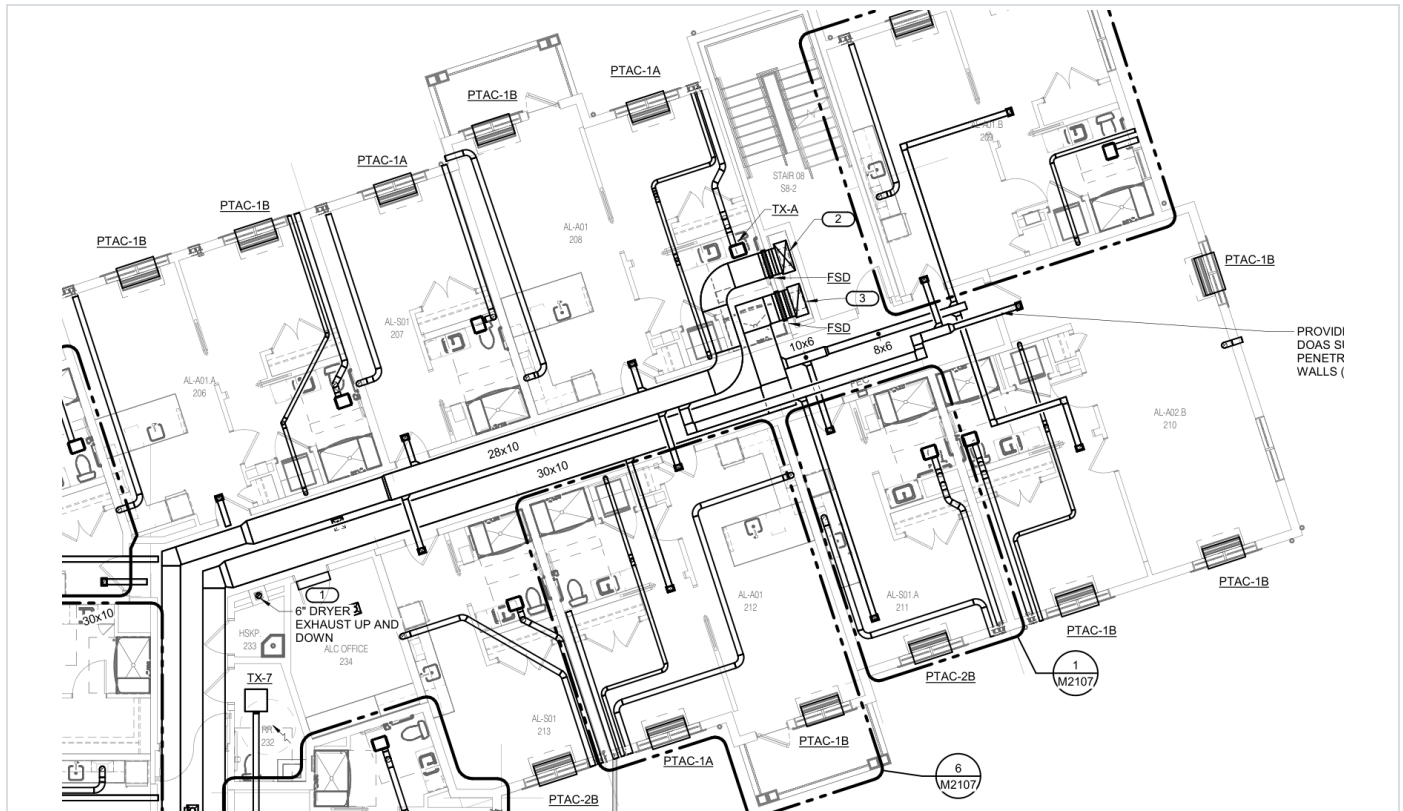
Proposed

- Apartment branches (6x4 to 8x6) under 100 sq in, 26 ga steel, no openings into the corridor, above the ceiling, with 12" steel sleeve: no damper required at the corridor fire partition (IBC 717.5.4 exception); plain fire dampers as fallback
- Fire / smoke dampers kept at shafts and smoke barriers

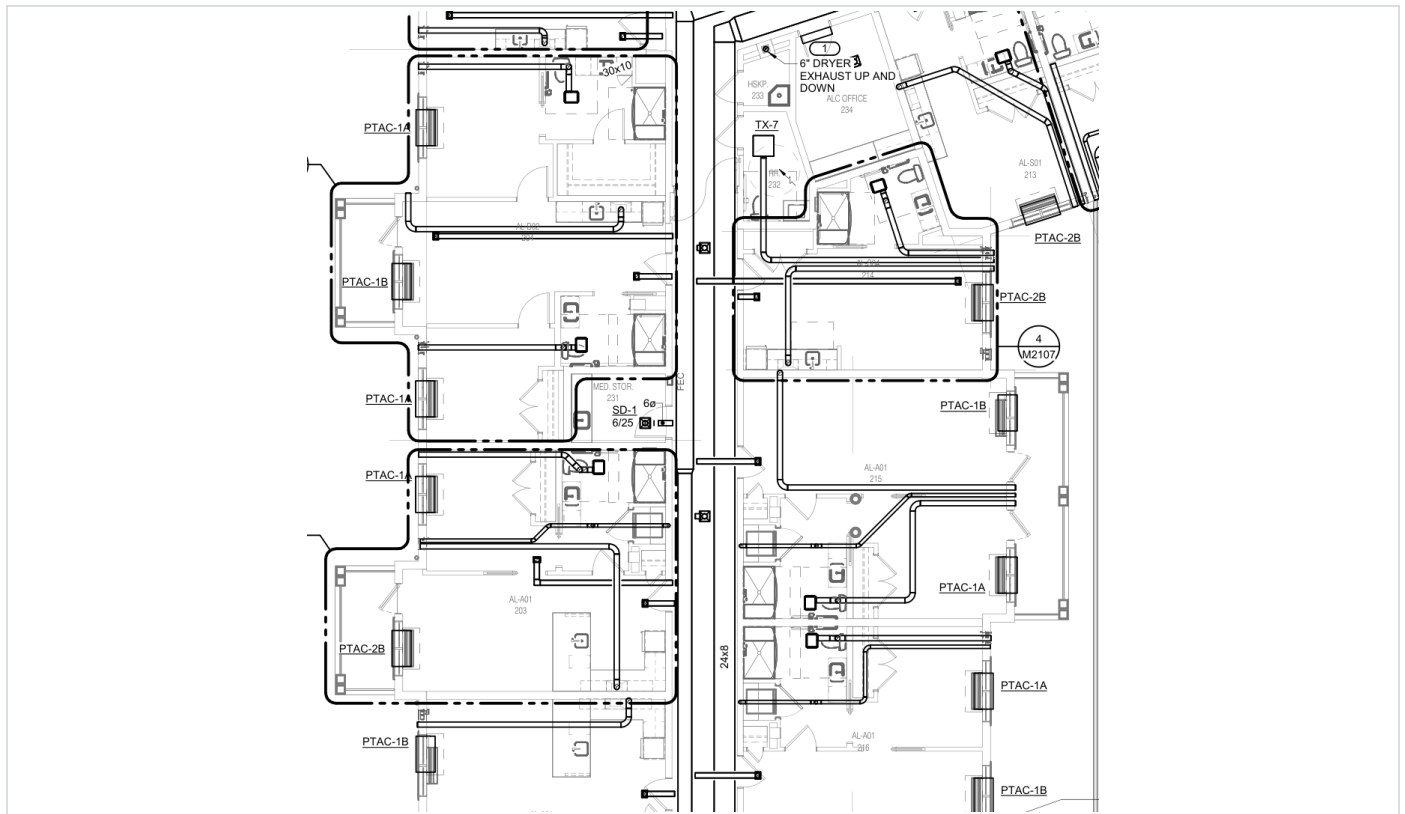
Benefits: Removes about 283 fire / smoke dampers with their power, fire alarm modules and access doors (mechanical, electrical and fire alarm scope).

Electrical impact: Reduces damper power circuits

Approvals / coordination: IMEG / HKS code analysis (AL Group I-1 smoke barriers), building department.



M2023 AL 2nd floor - DOAS corridor ducts, FSD at shaft and note for all units



M2023 AL 2nd floor - DOAS branches with FSD into each apartment

VE-08 Apartment bath exhaust to the DOAS energy recovery

Area: All apartments, IL and AL (TX-A)

As designed

- Each apartment bath has its own exhaust fan TX-A (75 CFM) ducted out the exterior wall
- DOAS return is taken from corridors and a few rooms; some AL unit types show a DOAS return grille RG-2, the floor plans connect the return trunk to only a few (to be clarified)

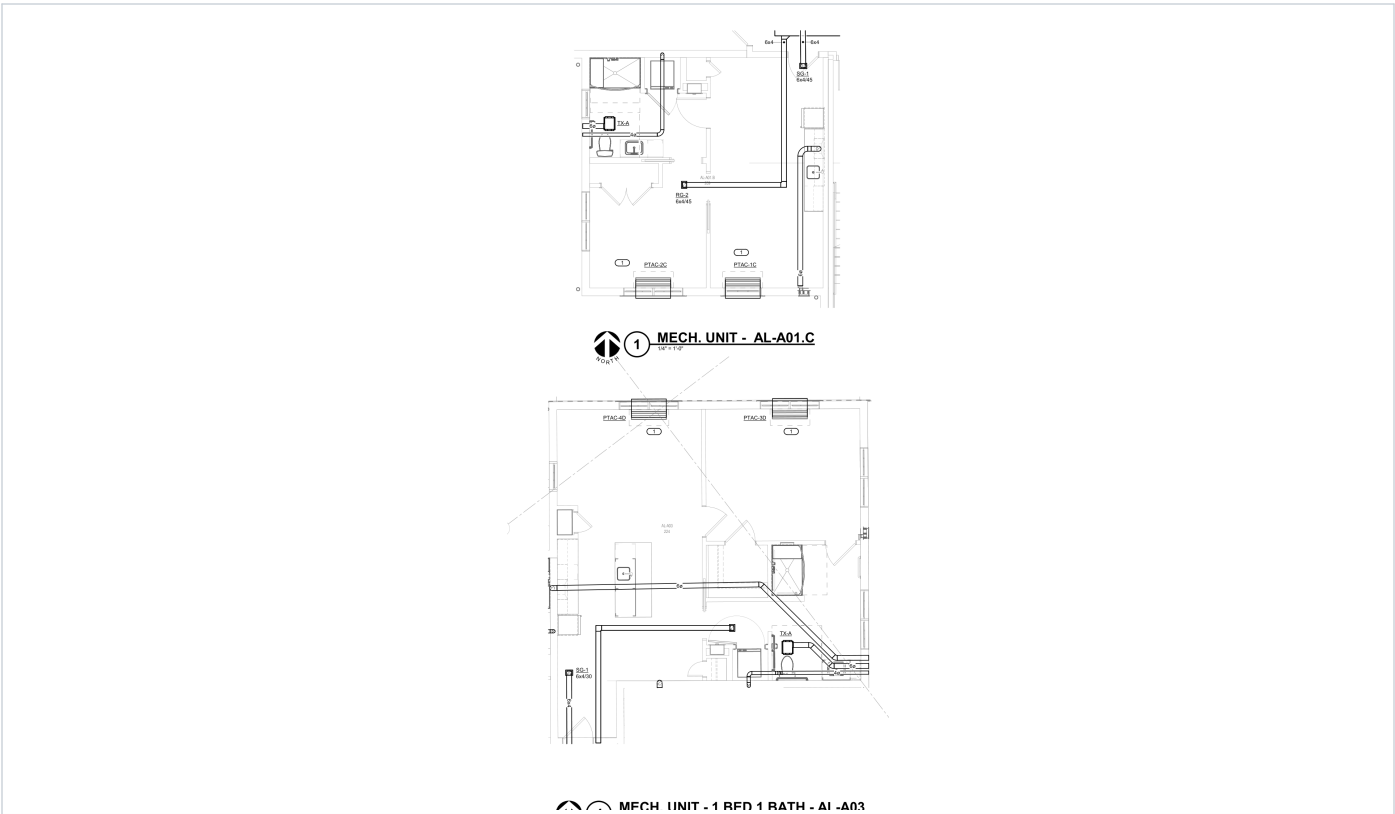
Proposed

- Duct apartment bath exhaust to the DOAS exhaust trunk in the corridor (as on Sunrise of Northport); the DOAS energy wheel recovers the exhaust heat
- Delete the individual TX-A fans, wall caps and their electrical circuits; delete RG-2 in the units

Benefits: Fewer wall penetrations and fans to maintain; better energy recovery and code compliance; the corridor return trunk serves every apartment.

Electrical impact: About -11 kW (fans deleted)

Approvals / coordination: IMEG; DOAS exhaust airflow check.



M2108 AL unit plans - TX-A bath exhaust to exterior, SG-1 / RG-2 DOAS grilles

VE-09 Controls: standalone equipment controls in lieu of BMS

Area: Building-wide

As designed

- Full BMS by the BMS contractor for boilers, pumps, heat exchangers, glycol, snow melt, VAV reheat, FCUs, cabinet heaters, rooftop units and DOAS

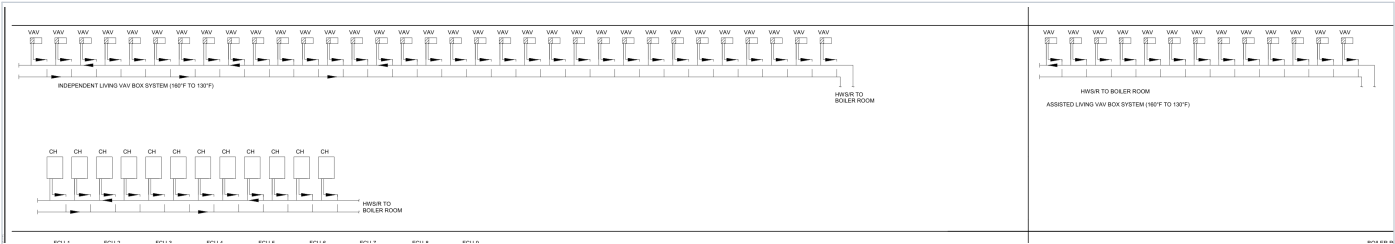
Proposed

- With VE-01 to VE-06 the hydronic sequences (about 120 to 150 points) are gone
- Rooftop units and DOAS on their factory controls with network cards for future monitoring; heaters, mini-splits and PTACs on standalone thermostats (as on other Sunrise communities)

Benefits: Large reduction or deletion of the BMS contract; simpler operation for the community staff.

Electrical impact: -

Approvals / coordination: Sunrise approval of the controls standard.



M5000 - BMS-controlled hydronic terminals eliminated by VE-01 to VE-05

Coordination items found during the review (RFI to follow)

- Basement piping sheets M2001 / M2003 show no boiler room, main or glycol piping.
- FCU counts do not match: schedule FCU-A quantity 20 (all data blank), M5000 FCU-1 to FCU-17, plans about 12.
- Pump schedule PP-1 / PP-2 blank. RTU-5 heating output 864 MBH is greater than its 600 MBH input.
- AL unit plans show DOAS return grilles RG-2 in some unit types; floor plans connect the return trunk to only a few rooms.

RFI-01 - Cooling and ventilation of equipment rooms (to IMEG)

Reference: mechanical GMP set dated 07.15.2026, sheets M2000, M2002, M2013, M6000. Not included in our VE proposals or pricing.

#	Room	Question
1	Electrical rooms B111 (Main Electrical), B120, B122 and B123	No cooling or ventilation is scheduled; B123 has a heating-only FCU. Transformer and switchgear heat is year-round. Please confirm the design heat load and provide cooling or thermostatic ventilation (ductless AC or exhaust / transfer fan) as required.
2	MDF B116 and IDF rooms (IDF 123 on 1st floor, IDF B134)	Please confirm the cooling source for the MDF / IDF rooms (24/7 load). A DX-FCU is shown near B115 / B116 on M2002 - confirm it serves the MDF and provide its schedule data.
3	Central Laundry B121	Heating-only FCU shown. Dryer heat and make-up air for the dryers: please confirm make-up air, exhaust and cooling / ventilation requirements for the laundry.
4	Mechanical B112 / B118 (with VE-01 boiler plant deleted)	Please confirm the ventilation requirement for the mechanical rooms once the boilers are deleted.

Response requested: design load and the system to be provided for each room, so it can be priced separately.